

3.5 Comparison Tests Worksheet

Directions: Determine if each of the following series converges or diverges. Use any technique from section 3.1 to section 3.5.

1. $\sum_{n=1}^{\infty} \frac{n}{2n^3 + 1}$

We first note that $\frac{n}{2n^3 + 1} < \frac{n}{2n^3} = \frac{1}{2n^2}$ for $n \geq 1$. We know that the series $\sum_{n=1}^{\infty} \frac{1}{2n^2} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n^2}$ is the a convergent p-series. Since both this series and the given series contain only non-negative terms we know, by the Comparison Test, that $\sum_{n=1}^{\infty} \frac{n}{2n^3 + 1}$ must also converge. ■

2. $\sum_{n=1}^{\infty} \frac{1}{2n+3}$

We first note that $\frac{1}{2n+3} < \frac{1}{2n} < \frac{1}{n}$ for $n \geq 1$. While we know that the series $\sum_{n=1}^{\infty} \frac{1}{n}$ is the Harmonic series and diverges we see that the Comparison Test won't allow us to reach any conclusion here since we will have shown that the series with larger terms is divergent. Therefore, we plan to use the Limit Comparison Test. Note that both the given series and the Harmonic series have only positive terms.

When then calculate that $\lim_{n \rightarrow \infty} \frac{1/n}{1/(2n+3)} = \lim_{n \rightarrow \infty} \frac{2n+3}{n} = \lim_{n \rightarrow \infty} \left(2 + \frac{3}{n} \right) = 2$. Since this value is positive and finite we know that both series have the same behavior. Since the Harmonic series diverges this means that $\sum_{n=1}^{\infty} \frac{1}{2n+3}$ must also diverge. ■

3. $\sum_{k=1}^{\infty} \frac{1}{(3+k)^{2/5}}$

We note that $\frac{1}{(3+k)^{2/5}} \geq \frac{1}{(k+k)^{2/5}} = \frac{1}{(2k)^{2/5}} = \frac{1}{2^{2/5} k^{2/5}}$ for $k \geq 3$. We see that the series

$\sum_{n=3}^{\infty} \frac{1}{2^{2/5} k^{2/5}} = \frac{1}{2^{2/5}} \sum_{n=3}^{\infty} \frac{1}{k^{2/5}}$ is a p-series with $p = \frac{2}{5} < 1$ and thus it diverges. Therefore, by the comparison

theorem, the series $\sum_{n=3}^{\infty} \frac{1}{(3+k)^{2/5}}$ diverges. So, $\frac{1}{4^{2/5}} + \frac{1}{5^{2/5}} + \sum_{n=3}^{\infty} \frac{1}{(3+k)^{2/5}} = \sum_{n=1}^{\infty} \frac{1}{(3+k)^{2/5}}$ diverges. ■

4.
$$\sum_{k=1}^{\infty} \frac{\cos\left(\frac{1}{k}\right)}{k^2}$$

We note that $0 < \cos\left(\frac{1}{k}\right) < 1$ for $k \geq 1$, thus $\frac{\cos\left(\frac{1}{k}\right)}{k^2} < \frac{1}{k^2}$. We see that the series $\sum_{n=1}^{\infty} \frac{1}{k^2}$ is a p -series with $p = 2 > 1$ and thus it converges. Therefore, by the comparison theorem, the series $\sum_{k=1}^{\infty} \frac{\cos\left(\frac{1}{k}\right)}{k^2}$ converges. ■

5. In the series $\sum_{n=1}^{\infty} \left(\frac{2^n}{x^n}\right)$, x stands for a real number. Depending on what number x is replaced with the given series may converge or diverge. Write an argument to find the values of x for which $\sum_{n=1}^{\infty} \left(\frac{2^n}{x^n}\right)$ converges.

We note that $\sum_{n=1}^{\infty} \left(\frac{2^n}{x^n}\right) = \sum_{n=1}^{\infty} \left(\frac{2}{x}\right)^n$, which is a geometric series with $r = \frac{2}{x}$. As long as $|r| < 1$ then the given series will converge. Therefore we must have: $|r| < 1 \Leftrightarrow \left|\frac{2}{x}\right| < 1 \Leftrightarrow -1 < \frac{2}{x} < 1$. This only happens when $\frac{x}{2} < -1$ or $\frac{x}{2} > 1$ which implies that $x < -2$ or $x > 2$. ■

6.
$$\sum_{k=1}^{\infty} \left(\frac{3}{2^k} - \frac{2}{3^k}\right)$$

We can rewrite the given series as follows: $\sum_{k=1}^{\infty} \left(\frac{3}{2^k} - \frac{2}{3^k}\right) = \sum_{k=1}^{\infty} \left[\frac{3}{2} \left(\frac{1}{2}\right)^{k-1} - \frac{2}{3} \left(\frac{1}{3}\right)^{k-1} \right]$. We see that we have a difference of two geometric series which are both convergence since $|r| < 1$ in both cases. Therefore we have:

$$\sum_{k=1}^{\infty} \left(\frac{3}{2^k} - \frac{2}{3^k}\right) = \sum_{k=1}^{\infty} \left[\frac{3}{2} \left(\frac{1}{2}\right)^{k-1} - \frac{2}{3} \left(\frac{1}{3}\right)^{k-1} \right] = \sum_{k=1}^{\infty} \frac{3}{2} \left(\frac{1}{2}\right)^{k-1} - \sum_{k=1}^{\infty} \frac{2}{3} \left(\frac{1}{3}\right)^{k-1} = \frac{\frac{3}{2}}{1 - \frac{1}{2}} - \frac{\frac{2}{3}}{1 - \frac{1}{3}} = 3 - 1 = 2$$

■

7. $\sum_{k=1}^{\infty} \left(2 - \frac{1}{k}\right)$

We note that $\lim_{k \rightarrow \infty} \left(2 - \frac{1}{k}\right) = 2$. Therefore, by the test for divergence, the series $\sum_{k=1}^{\infty} \left(2 - \frac{1}{k}\right)$ diverges. ■

8. $\sum_{n=1}^{\infty} \frac{4^{n+1}}{3^n - 2}$

We first note that $\frac{4^{n+1}}{3^n - 2} > \frac{4^{n+1}}{3^n} = 4 \left(\frac{4}{3}\right)^n$ for $n \geq 1$. We know that the series $\sum_{n=1}^{\infty} 4 \left(\frac{4}{3}\right)^n$ is a geometric series with $r = \frac{4}{3}$ and since $|r| \geq 1$ this series diverges. Since both this series and the given series contain only non-negative terms we know, by the Comparison Test, that $\sum_{n=1}^{\infty} \frac{4^{n+1}}{3^n - 2}$ must also diverge. ■

9. $\sum_{n=1}^{\infty} \frac{n+2}{(n+1)^3}$

We first note that $\frac{n+2}{(n+1)^3} < \frac{n+2}{n^3} = \frac{1}{n^2} + \frac{2}{n^3}$ for $n \geq 1$. We know that the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ & $\sum_{n=1}^{\infty} \frac{2}{n^3}$ are convergent p-series and so the series $\sum_{n=1}^{\infty} \left(\frac{1}{n^2} + \frac{2}{n^3}\right)$ is also convergent. Since both this series and the given series contain only non-negative terms we know, by the Comparison Test, that $\sum_{n=1}^{\infty} \frac{n+2}{(n+1)^3}$ must also converge. ■

10. $\sum_{k=1}^{\infty} \frac{\arctan(k)}{1+k^2}$

Let $f(x) = \frac{\arctan(x)}{1+x^2}$. We see that $f(x)$ is a continuous and positive function on $[1, \infty)$. We also note

that $f'(x) = \frac{1-2x \arctan(x)}{(1+x^2)^2}$ and $1-2x \arctan(x) < 0$ for $x \geq 1$ since $\frac{\pi}{4} \leq \arctan(x) \leq \frac{\pi}{2}$ for $x \geq 1$.

Therefore we see that $f'(x) < 0$ for $x \geq 1$, which means that $f(x)$ is decreasing on $[1, \infty)$. So we can use the integral test, and we consider the following:

$$\int_1^{\infty} \frac{\arctan(x)}{1+x^2} dx = \lim_{t \rightarrow \infty} \left(\int_1^t \frac{\arctan(x)}{1+x^2} dx \right) = \lim_{t \rightarrow \infty} \left(\int_{\arctan(1)}^{\arctan(t)} u du \right) = \lim_{t \rightarrow \infty} \left(\frac{u^2}{2} \Big|_{\pi/4}^{\arctan(t)} \right) = \lim_{t \rightarrow \infty} \left(\frac{[\arctan(t)]^2}{2} - \frac{\pi^2}{32} \right) = \frac{\pi^2}{8} - \frac{\pi^2}{32} = \frac{3\pi^2}{32}$$

Here we used the substitution: $u = \arctan(x)$ which implies $du = \frac{1}{1+x^2} dx$.

Therefore, by the integral test, since $\int_1^{\infty} \frac{\arctan(x)}{1+x^2} dx$ converges, the series $\sum_{k=1}^{\infty} \frac{\arctan(k)}{1+k^2}$ is convergent. ■

11. $\sum_{n=1}^{\infty} \frac{n^2-5n}{n^3+n+1}$

We first note that $\frac{n^2-5n}{n^3+n+1} < \frac{n^2}{n^3+n+1} < \frac{n^2}{n^3} = \frac{1}{n}$ for $n \geq 1$. While we know that the series $\sum_{n=1}^{\infty} \frac{1}{n}$ is the

Harmonic series and diverges we see that the Comparison Test won't allow us to reach any conclusion here since we will have shown that the series with larger terms is divergent. Therefore, we plan to use the Limit Comparison Test. Note that both the given series and the Harmonic series have only positive terms. When then calculate that

$$\lim_{n \rightarrow \infty} \frac{\left(\frac{n^2-5n}{n^3+n+1} \right)}{\left(\frac{1}{n} \right)} = \lim_{n \rightarrow \infty} \frac{n^3-5n^2}{n^3+n+1} = \lim_{n \rightarrow \infty} \frac{n^3-5n^2}{n^3+n+1} \cdot \frac{1/n^3}{1/n^3} = \lim_{n \rightarrow \infty} \frac{1-5/n}{1+1/n^2+1/n^3} = 1. \text{ Since this value is}$$

positive and finite we know that both series have the same behavior. Since the Harmonic series diverges

this means that $\sum_{n=1}^{\infty} \frac{1}{2n+3}$ must also diverge. ■